



EcoRoute Agent

AI-Driven Supply Chain Sustainability

Capstone Project for Lenovo Leap Next Gen Scholar Program

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SDG 12

Responsible Consumption & Production



Hidden Cost of Last-Mile Delivery

Last-mile delivery is the least efficient leg of the supply chain — fragmented routes, partial loads, and repeat trips quietly drive up fuel use and emissions per parcel delivered.



Why Traditional Software Fails Small Business

Enterprise logistics platforms are built for scale — high license costs, long setup times, and complex configuration put them out of reach for small and local operators.

Beyond Traditional Automation

TRADITIONAL SCRIPTS

Rigid “if / then” rules that only work for the exact cases they were written for.

- Fixed decision trees
- Breaks on unseen scenarios
- No memory or reasoning
- Manual updates required

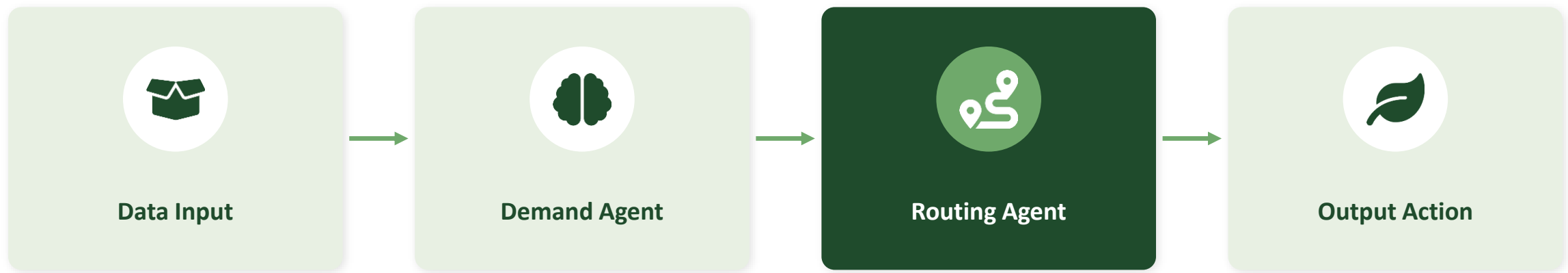


OUR AGENTIC SYSTEM

Reasons through problems, prioritizes tasks, calls external tools, and adapts as conditions change in real time.

- Dynamic reasoning & prioritization
- Uses external tools via APIs
- Adapts to changing routes / stock
- Learns from new data on the fly

Under the Hood



Built Entirely in Python

Agent orchestration frameworks paired with core data-science libraries power reasoning, tool calls, and route optimization end to end.



Scalability & Sustainability Goals



Measurable Carbon Reduction

A quantifiable drop in delivery-related carbon footprint for local and small businesses adopting the agent.



IoT Integration for Real-Time Tracking

Future integration with IoT sensors will bring live fleet, cargo, and environmental data directly into the agent's decision loop.

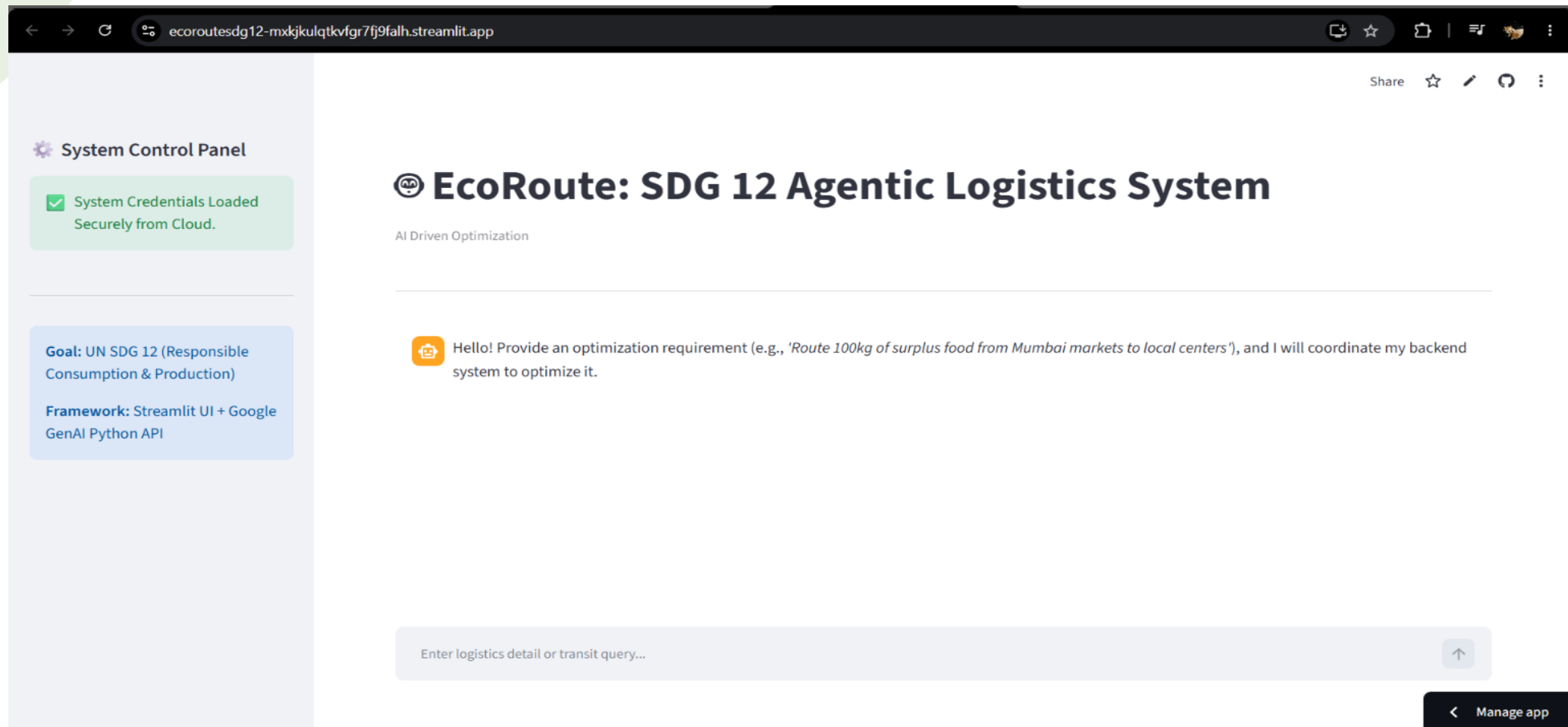


Built to Scale with Sustainability at the Core

A lightweight, low-cost architecture designed so small operators can grow their fleet without growing their environmental footprint.

Screenshots

(The working)



Thank You

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Screenshots

(The working)

🤖 EcoRoute: SDG 12 Agentic Logistics System

AI Driven Optimization



Hello! Provide an optimization requirement (e.g., *'Route 100kg of surplus food from Mumbai markets to local centers'*), and I will coordinate my backend system to optimize it.



Route 150kg of perishable dairy and fresh produce from wholesale markets in Dadar to 3 community distribution centers across South Mumbai. High ambient humidity, tight delivery windows, and standard diesel vans are currently allocated.



🔄 EcoRoute system analyzing parameters and generating green transit alternatives...

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Screenshots

(The working)

Stage 1: Supply Chain Waste Analyzer (Persona: Supply Chain Waste Analyzer)

Cargo Analysis: 150kg Perishable Dairy and Fresh Produce

The input cargo presents a high-risk profile for material waste and deterioration, particularly under the specified conditions.

Deterioration Vectors & Waste Drivers:

1. Temperature Abuse & Spoilage:

- **Dairy:** Highly susceptible to bacterial growth and spoilage if not maintained within a strict cold chain (typically 0-4°C). Fluctuations accelerate souring and curdling.
- **Fresh Produce:** Varies by type, but most produce requires cool temperatures to slow respiration, ripening, and enzymatic degradation. High temperatures accelerate wilting, softening, and microbial decay.
- **Current Allocation Impact:** The use of "standard diesel vans" strongly implies a lack of refrigeration. This is the single most critical deterioration vector, guaranteeing significant spoilage of both dairy and produce, especially in a warm, humid environment like Mumbai.

2. High Ambient Humidity:

- **Impact:** Mumbai's high humidity will exacerbate spoilage. It promotes mold and bacterial growth on produce surfaces and can lead to condensation within the vehicle if temperatures fluctuate, further compromising dairy packaging and product integrity. Produce will wilt faster.

3. Physical Damage:

- **Vulnerability:** Both dairy products (e.g., cartons, bottles) and fresh produce (e.g., soft fruits, leafy greens) are delicate.

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Screenshots

(The working)

- **Tight Delivery Windows Impact:** Rushed loading/unloading and potentially aggressive driving to meet deadlines can lead to bruising, crushing, and breakage, rendering products unsellable or significantly reducing their shelf life. Improper stacking in an unrefrigerated van can also cause damage and restrict airflow, accelerating spoilage.

4. Time Sensitivity:

- **Perishability:** Dairy and fresh produce have short shelf lives. "Tight delivery windows" are essential, but without proper cold chain management, even quick transit times can be too long if the product is exposed to adverse conditions. Delays due to traffic or operational inefficiencies will directly translate to increased waste.

UN SDG 12 (Responsible Consumption and Production) Relevance:

The current setup directly undermines several targets under SDG 12:

- **Target 12.3: Halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses.**
 - **Analysis:** The lack of refrigerated transport for highly perishable goods in a challenging climate will inevitably lead to substantial post-harvest losses due to spoilage. This directly contradicts the goal of reducing food loss in the supply chain. The 150kg of cargo is at high risk of becoming waste before reaching distribution centers.
- **Target 12.4: Achieve the environmentally sound management of chemicals and all wastes throughout their life cycle.**
 - **Analysis:** While not chemical waste, the generation of significant food waste requires disposal, which often involves landfilling (producing methane, a potent greenhouse gas) or energy-intensive composting. Preventing this waste through proper logistics is a more environmentally sound approach.

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Conclusion

The road ahead



A Practical Path to SDG 12

EcoRoute Agent shows that responsible consumption and production isn't limited to large enterprises — it's achievable for small logistics operators too.



Agentic AI Outperforms Static Scripts

By reasoning, prioritizing, and adapting in real time, the agent handles the messiness of real-world logistics that rigid automation cannot.



Clear Next Steps

IoT sensor integration, expanded API partnerships, and pilot deployments with local businesses will validate impact at scale.

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Working Links

Link to the GitHub Repository :

https://github.com/Kashmira736/Ecoroute_sdg12

Link to the Streamlit App :

<https://ecoroutesdg12-mxkjkulqtkvfgr7fj9falh.streamlit.app/>

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